

Understanding Summation Notations

A Crystal-Clear Guide to $\sum_{i=1}^n$ and $\sum_{i \in I_j}$ in Machine Learning & Decision Trees

When studying advanced machine learning algorithms—especially tree-based gradient boosting models like XGBoost—you frequently encounter two vital summation notations. Let's break them down step by step in the simplest way possible.

1. What is $\sum_{i=1}^n f_i$?

The standard summation symbol aggregates values across an entire dataset.

$$\sum_{i=1}^n f_i$$

- **i = 1** (Lower limit) → Where the summation starts (first sample index).
- **n** (Upper limit) → Where the summation ends (total sample index).
- **i** → The index or serial number of the training sample.
- **n** → The total number of training samples in the dataset.

Concrete Example

Suppose our dataset contains **5** training samples (**n = 5**). Expanding the summation gives:

$$\sum_{i=1}^5 f_i = f_1 + f_2 + f_3 + f_4 + f_5$$

2. What does $i \in I_j$ mean?

When a decision tree splits, training samples are distributed into various terminal leaf nodes. Consider the formula:

$$H_j = \sum_{i \in I_j} f_i$$

- **j** → The leaf number (e.g., Leaf 1, Leaf 2).
- **I_j** → The set of sample indices belonging to leaf **j**.

- $i \in I_j \rightarrow$ Means "sum only over those sample indices that reside inside leaf j ."

Step-by-Step Example

Suppose we have **6** total samples in our dataset: $i = 1, 2, 3, 4, 5, 6$ ($n = 6$).

After a tree split, samples are distributed into two leaves:

- **Leaf 1** ($j = 1$) receives samples x_1, x_3, x_5 & implies; $I_1 = \{1, 3, 5\}$
- **Leaf 2** ($j = 2$) receives samples x_2, x_4, x_6 & implies; $I_2 = \{2, 4, 6\}$

Applying the formula for second derivatives (Hessians) H_j :

- For Leaf 1:

$$H_1 = \sum_{i \in I_1} f_i = f_1 + f_3 + f_5$$

- For Leaf 2:

$$H_2 = \sum_{i \in I_2} f_i = f_2 + f_4 + f_6$$

3. What is i and f_i ?

The index i is the serial number of each sample in the dataset:

Index ()	Sample Representation & Meaning
1	x_1 (Sample 1 $\rightarrow f_1$ is its second derivative / Hessian)
2	x_2 (Sample 2 $\rightarrow f_2$ is its second derivative / Hessian)
3	x_3 (Sample 3 $\rightarrow f_3$ is its second derivative / Hessian)
4	x_4 (Sample 4 $\rightarrow f_4$ is its second derivative / Hessian)
5	x_5 (Sample 5 $\rightarrow f_5$ is its second derivative / Hessian)
6	x_6 (Sample 6 $\rightarrow f_6$ is its second derivative / Hessian)

4. Summary Comparison

Notation Expression	Core Meaning
$\sum_{i=1}^n$	Sum across all n training samples in the entire dataset.
$\sum_{i \in I_j}$	Sum only over the subset of samples belonging to leaf j .
$G_j = \sum_{i \in I_j} g_i$	The total sum of gradients (g_i) inside leaf j .
$H_j = \sum_{i \in I_j} f_i$	The total sum of Hessians / second derivatives (f_i) inside leaf j .